

Rewriting Expressions with Distributive Structure

Distributing a factor, taking a common factor out, and choosing the form that answers the question

Directions

- Every quantity below is stated with its unit. Give your answer in the unit the problem names.
 - “Without brackets” means expanded form; “as a product” means factored form.
 - Show the distributing or factoring step, not just the final expression.
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1. A rectangular dog run measures $(2x + 9)$ m along its long side and 4 m along its short side. Write its perimeter, in metres, as an expression with no brackets.

Answer: _____ m

2. A school shop sells one notebook for $(x + 3)$ dollars. A teacher buys 15 identical notebooks. Write the total cost, in dollars, as an expression with no brackets.

Answer: _____

3. A hallway floor has area $(6x + 30)$ square metres and a constant width of 6 m. Its area is width $\times$ length. Find the length of the hallway, in metres.

Answer: _____ m

4. A club orders class sets of books. Plan A charges $8(x + 5)$ dollars and Plan B charges $8x + 40$ dollars, where x is the number of books a class needs.

(a) Rewrite Plan A without brackets to confirm the two plans always charge the same.

Answer: _____

(b) One printed form shows the price per book and the 5 spare books; the other shows a flat 40-dollar charge as its own term. Say which is which and explain what each form makes visible.

Answer: _____

5. A patio is a rectangle measuring $(x + 6)$ m by $(x + 2)$ m. Write its area, in square metres, in standard form.

Answer: _____ m^2

6. A crew has painted $(9x^2 + 15x)$ square metres of wall. Write that area as a product of the greatest common monomial factor and one remaining factor.

Answer: _____

7. A stall sells x bunches of flowers. It takes in $40x$ dollars and its costs are $5(3x + 20)$ dollars. Profit is income minus costs. Write the profit, in dollars, as an expression with no brackets.

Answer: _____

8. A field is $(3x + 12)$ m long. It is divided along its length into 3 strips of equal length.

(a) Find the length of one strip, in metres.

Answer: _____ m

(b) Explain how writing the field length in the factored form $3(x + 4)$ makes that division something you can read off rather than calculate.

Answer: _____

9. A border is built from three straight runs of edging: 3 runs of $(2x + 5)$ m, 4 runs of $(x - 2)$ m, and then 2 runs of $(x + 1)$ m are removed again. Write the total length of edging left, in metres, as an expression with no brackets.

Answer: _____ m

10. A rectangular metal sheet measures $(3x + 6)$ cm by $(x + 4)$ cm.

(a) Write the area of the sheet in standard form, in square centimetres.

Answer: _____ cm^2

(b) Write the same area in fully factored form.

Answer: _____

(c) The workshop needs to know that the area is always a whole number of square centimetres divisible by 3 whenever x is a whole number. Say which of your two forms shows that immediately, and explain why.

Answer: _____